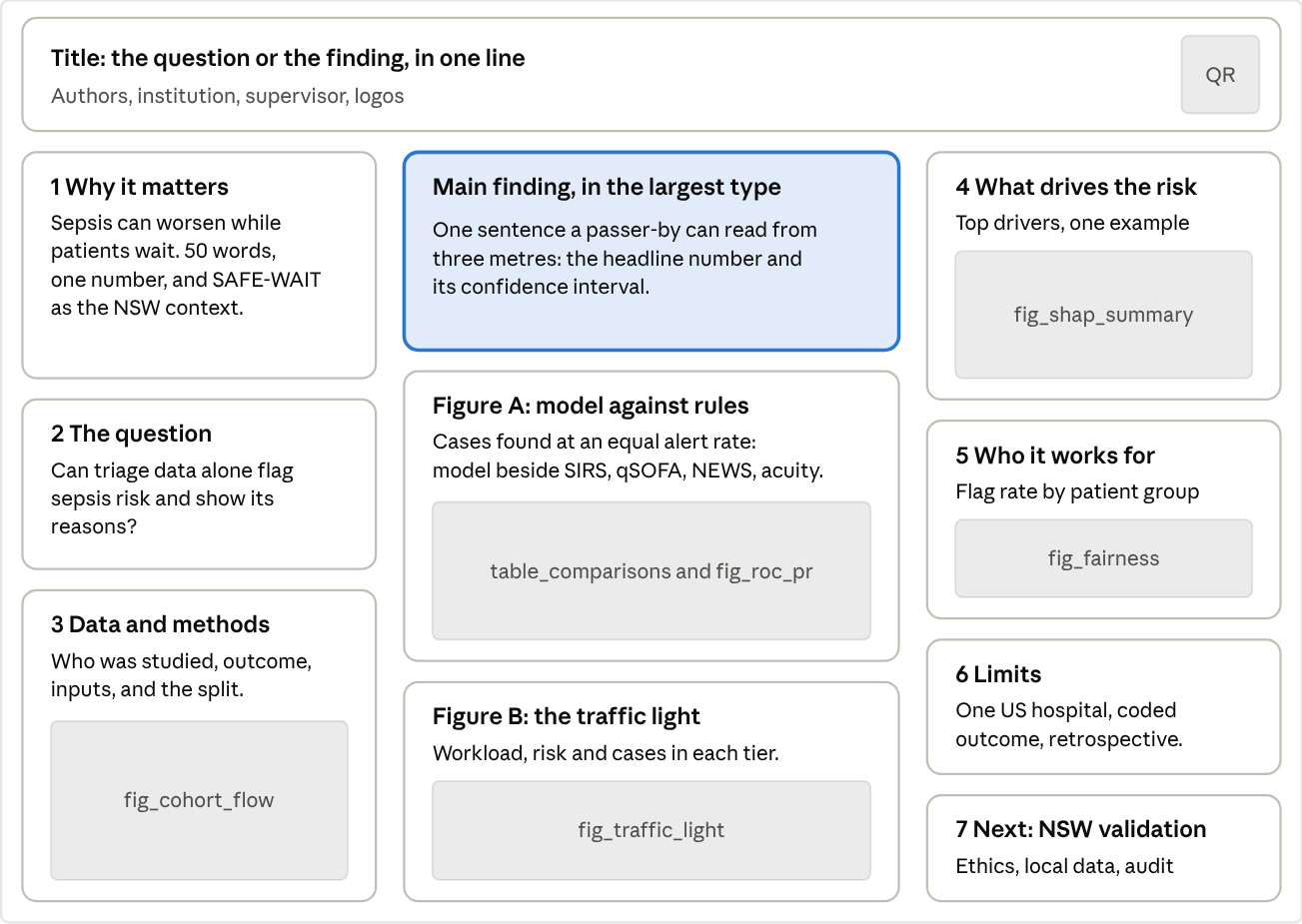
Poster Competition Kit — Sepsis Risk at ED Triage

Oct 2, 2026 · @Rupesh Ghimire

Poster layout and content

The poster is built around one sentence in the centre, with the reasons to believe it on either side; a reader moves left, centre, right.

Poster layout: the finding in the centre, evidence on either side



poster wireframe · landscape, 3 columns, 10 panels

The grey blocks are figures the analysis code writes into the `outputs` folder. The layout is landscape; for a portrait poster, stack the three columns as left, centre, right from top to bottom and keep the main finding directly under the title.

Panel	Content	Word budget
Title	The finding or the question in one line. Authors, institution, supervisor, QR code	20
1 Why it matters	Sepsis can worsen in the waiting room. One statistic. SAFE-WAIT as the NSW context	50
2 The question	One sentence, then the five things measured as a short list	40
3 Data and methods	Cohort flow figure. Outcome, inputs and split, two lines each. Three models, four rule-based comparators	80
Main finding	One sentence with the headline number and its confidence interval	30
Figure A	Share of sepsis cases found by the model and by each rule at the same alert rate. Small table of AUROC and AUPRC underneath	40
Figure B	Traffic-light figure: share of arrivals, sepsis rate and share of cases in each tier. Label: research prototype	40
4 What drives the risk	SHAP summary of the top 10 features. One invented patient with three reasons	50
5 Who it works for	Subgroup figure. One sentence on the largest gap	40
6 Limits	Four bullet points	40
7 Next: NSW validation	Three steps: ethics approval, local data, equity audit including Aboriginal and Torres Strait Islander patients	40
Footer	Three key references, data citation, acknowledgements	30

Choosing the main finding

Write it only after the test set is opened. Use the matched-alert comparison if it is clear, because it speaks to workload. A template: "Using triage information alone, the model found [x] in 100 sepsis cases; the standard screening rule, flagging the same number of patients, found [y]."

Open question: the competition's required poster size and orientation. Set the canvas to that size before placing anything.

Design and print checklist

A judge gives a poster about ten seconds before deciding whether to stop, so the design job is to make one finding readable from three metres and everything else easy to find.

Type sizes for an A0 poster (scale by 0.7 for A1)

Element	Size	Notes
Title	85 to 100 pt	One line if possible, sentence case
Main finding	60 to 72 pt	The largest text after the title
Section headings	44 to 54 pt	Numbered, so the reading order is obvious
Body text	28 to 32 pt	Never smaller than 24 pt
Figure labels and captions	22 to 26 pt	Re-export figures at this size; do not stretch small images
References and acknowledgements	18 to 20 pt	Short forms are fine

Rules

- 500 words or fewer on the whole poster, excluding figure labels.
- One typeface, two weights. A plain sans-serif such as Inter, Source Sans or Arial.
- One accent colour for the model, grey for comparators, and red, amber and green only for the three tiers.
- Every figure has a one-line caption that states its finding, not its type.

- Left-align body text. Leave at least 25 mm of margin and clear space between panels.
- Put the NSW relevance and the limits in plain view. Judges trust a poster that states its limits.

Figures: use the vector files

The pipeline saves each figure as PDF and 300 dpi PNG. Place the PDF versions so text stays sharp at print size. In PowerPoint, set the slide to the print size first (A0 is 1189 by 841 mm), then insert.

Before printing

- [ ] Poster size and orientation match the competition rules
- [ ] Every number matches `results_sentences.md` from the final run
- [ ] No stand-in watermark anywhere
- [ ] No patient-level data, and the individual example is an invented patient
- [ ] "Research prototype, not for clinical use" appears on the traffic-light panel
- [ ] SAFE-WAIT is described as motivation and context only
- [ ] QR code tested on two phones; it opens the code and the reporting checklist
- [ ] Institution and supervisor acknowledged; data source cited
- [ ] Printed at 100% on one A3 sheet and read from arm's length as a proof
- [ ] Print order placed at least three working days ahead

One-page handout

Print 30 copies: the humanised abstract, the traffic-light table, the QR code, and team contact details. Judges take handouts back to the scoring room.

Pitch scripts

Three lengths cover every situation at a poster session: a passer-by, a judge on a timer, and a judge who stays. Brackets are filled from the final results; say the numbers from memory.

30 seconds: the passer-by (about 70 words)

> Picture sixty people in an emergency waiting room. One has an infection that is turning into sepsis, and nobody can tell yet. We asked how much the first few minutes of triage information can reveal. Using only vital signs, age and the reason for the visit, our model found [x]% of sepsis cases. The standard screening rule, flagging the same number of patients, found [y]%. And every flag comes with its reasons.

2 minutes: the timed judge (about 280 words)

> Sepsis can go from a treatable infection to organ failure in hours, and it often starts quietly. In a crowded emergency department the risk is the patient who looked fine at triage and is getting worse in the waiting room.
>
> NSW Health is tackling this with SAFE-WAIT, an AI tool being trialled across five Western Sydney hospitals from 2027. That told us the problem matters here. Our question was narrower: what can triage information alone achieve, and can the model explain itself?
>
> We used [N] de-identified emergency visits from a public research database. The model sees only what a triage nurse records: vital signs, age, the presenting complaint, how the patient arrived, and past visits. No blood tests. We compared three models with the screening rules used today, on patients the models had never seen.
>
> Three results. First, at the same number of alerts as the standard rule, the model found [x]% of sepsis cases and the rule found [y]%. Second, the top [5]% of patients by risk held [x]% of all sepsis cases, about one case in every [n] reviews. Third, the reasons make clinical sense: [feature 1], [feature 2] and [feature 3] drive the score, and each patient's flag shows its own top three.
>
> We also checked who it works for. Performance was [similar across / weaker for] [groups], which we report openly.
>
> The limits are clear. This is one American hospital, and sepsis was defined by diagnosis codes. It is a research prototype. The next step is testing on de-identified NSW data with ethics approval, and we have mapped that pathway.

5 minutes: the judge who stays

Time	Point at	Say
0:00 to 0:40	Panel 1	The waiting-room story, then SAFE-WAIT as proof that NSW cares about this
0:40 to 1:20	Panels 2 and 3	The question, the data, and the rule that the model sees triage information only
1:20 to 2:30	Centre figures	The matched-alert comparison first, then the traffic-light table. Give workload numbers before AUROC
2:30 to 3:15	Panel 4 and the laptop	What drives risk. Run one invented patient through the demo and read out the three reasons
3:15 to 3:50	Panel 5	The subgroup audit, including any gap you found
3:50 to 4:20	Panel 6	Limits, said plainly and without apology
4:20 to 5:00	Panel 7	The NSW validation pathway, then one question back to the judge: "What would you want to see before trusting this in your department?"

Delivery notes

- Open with the patient, not the algorithm.
- Say "found [x] in 100 cases", not "sensitivity of 0.x". Translate every metric once.
- Stand to the side of the poster and point at the figure you are describing.
- Stop after the finding and let the judge ask. A conversation scores better than a recital.

Likely judge questions with model answers

These fourteen questions cover what clinical and technical judges usually probe. Each answer is short on purpose: answer, give one reason, stop.

Question	Model answer
Why did you not use NSW data?	NSW hospital data needs ethics and custodian approval, which takes months. We built and tested the method on public data first, so the NSW study can start from working, checked code.
How is this different from SAFE-WAIT?	SAFE-WAIT is a deployed clinical program evaluated on time to antibiotics. Ours is an open, reproducible study of what triage data can predict, with explanations and a subgroup audit. We do not use or test their model.
Why would a US dataset tell us anything about NSW?	It tells us whether the method works and what to measure. It does not tell us performance in NSW. That is why local validation is our stated next step.
How did you define sepsis?	By diagnosis codes in the ED record or the linked admission. Codes under-count sepsis and have no onset time, which we list as a limitation. We checked the result against a Sepsis-3 definition in patients who went to ICU.
Is the model just learning what the nurse already knows?	We tested that. The primary model excludes the nurse's acuity score. Adding it changed AUROC from [x] to [y], and the model [did / did not] outperform acuity alone.
Why XGBoost and not deep learning?	The data are about 60 columns of structured values, where boosted trees are strong and explainable. We also show logistic regression came within [d] of it, which matters for transparency.
How do you know there is no data leakage?	Three safeguards. The code blocks any variable recorded after triage. Patients are split so no one is in both training and testing. And our result sits within the range published for this database.

Question	Model answer
AUROC is high, but sepsis is rare. How many false alarms?	In the high tier, about one in [n] flagged patients had sepsis. We report alerts per 100 arrivals for this reason, because AUROC hides workload.
How did you handle class imbalance?	We did not oversample. Published work shows it distorts predicted risks. We handled it through the choice of cut-off, and reported precision-recall and calibration.
Is it fair?	We audited by sex, age, recorded race, arrival mode and missing data. The largest gap was [group], at [x] points. In NSW the audit must cover Aboriginal and Torres Strait Islander patients, which this dataset cannot do.
Can SHAP explanations be trusted?	They describe what the model used, not what causes sepsis. We cross-checked the ranking against two other methods and asked a clinician whether the drivers were plausible.
What would it take to use this in a hospital?	Local validation, a prospective silent trial, the NSW AI Assessment Framework, a check against medical device rules, and nurse-led workflow design. We are at step zero of that list and say so.
What would you do differently?	Use a clinical sepsis definition with onset time for everyone, and add the triage nurse's free-text note, which other studies found helpful.
Did you use AI tools?	Answer honestly and specifically: which tools, for what, and that every analysis choice and number was checked by the team. Never with patient-level data.

Each answer assumes the matching step in the project plan was done. Two are easy to skip: the Sepsis-3 check and the clinician review. If a step was skipped, change the answer to say so.

If you do not know the answer

Say: "We did not test that. Here is how we would." Then describe the check in one sentence. Judges mark honesty and method above certainty.

Rehearsal plan and day-of checklist

Three rehearsals in the final week are enough, provided the third is in front of someone outside the team.

Rehearsal	When	Audience	Goal
1	Six days before	Team only	Everyone can give the 30-second and 2-minute pitch without notes
2	Four days before	Supervisor or a clinician	Stress-test the clinical claims; collect five hard questions
3	Two days before	Someone with no health or data background	Check that the story is clear; time every version

Who says what

- One person leads each pitch. Others add only when asked.
- Agree who takes clinical questions, who takes modelling questions, and who runs the demo.
- Everyone can answer the first three questions in the table above.

Bring on the day

- [ ] Printed poster in a tube, plus fixings (pins or Velcro: check which the venue uses)
- [ ] A3 backup print and a PDF on a USB stick and a phone
- [ ] Laptop with the demo running offline on the real trained model, plus a charger
- [ ] 30 handouts
- [ ] Results sheet for the team: the six headline numbers with confidence intervals
- [ ] Water, and a plan for who covers the poster during breaks

At the poster

- Arrive 30 minutes early and check the poster reads from three metres.
- Greet, give the 30-second pitch, then ask: "Would you like the two-minute version or shall I answer questions?"
- Write down every question a judge asks. They become the discussion section of the paper.

After the session

- Send the supervisor the question list and the judges' feedback the same day.
- Decide within a week whether to pursue the NSW validation, and who drafts the ethics application.